

National Higher School of Mathematics

Preparatory Cycle
First Year

Academic Year: 2024/2025
Course: Algebra 1

Worksheet N°5

Polynomials and rational fractions

The starred exercises are to be done in tutorial sessions

Exercise 1. ★ For each of the following statements, determine whether the statement is true or false and justify your answer.

1. Let $P \in \mathbb{R}[X]$ and $z_0 \in \mathbb{C}$. If $P(z_0) = 0$, then $P(\overline{z_0}) = 0$.
2. The polynomial $X^4 + 1$ is irreducible in $\mathbb{R}[X]$.
3. The polynomial $X^3 - 2$ is irreducible in $\mathbb{Q}[X]$.
4. If two polynomials in $\mathbb{C}[X]$ have the same roots, then they are equal.
5. Every polynomial in $\mathbb{R}[X]$ of odd degree has at least one real root.
6. Let $P \in \mathbb{Q}[X]$ be a polynomial of degree n . The irreducible factors of P over $\mathbb{Q}$ are of degree 1 or 2.
7. The polynomial $X^4 + 1$ is irreducible in $\mathbb{Z}/2\mathbb{Z}[X]$.
8. In the ring $\mathbb{Z}/3\mathbb{Z}[X]$, the polynomial function associated to the polynomial $X^3 - X$ is the zero function.
9. In the ring $\mathbb{Z}/4\mathbb{Z}[X]$, the polynomial $(1 + 2X)$ is invertible.
10. If P is a real polynomial such that $\forall x \in \mathbb{R}, P(x) = P(\cos x)$, then $P = 0$.
11. Let $\mathbb{K}$ be a field and let $P \in \mathbb{K}[X]$. If $P(\alpha) = 0$ for all $\alpha \in \mathbb{K}$, then P is the zero polynomial.
12. If R is a commutative ring with $1 \neq 0$, and $P, Q \in R[X]$ are polynomials of degree two and three respectively, then the degree of PQ is five.
13. The fraction $F = \frac{1}{X}$ does not have a primitive in $\mathbb{R}(X)$.
14. There does not exist a rational function F such that $F^2 = X$.

Exercise 2. ★

1. Determine all polynomials P with coefficients in $\mathbb{R}$ such that

$$P(X^2 + 1) = P(X).$$

2. Determine all polynomials P with coefficients in $\mathbb{C}$ such that

$$P(X^2) = X^2(X^2 + 1)P(X).$$

3. Determine all polynomials P with coefficients in $\mathbb{C}$ such that

$$P' \text{ divides } P.$$

4. Determine all polynomials with coefficients in $\mathbb{C}$ such that

$$P(\mathbb{C}) \subset \mathbb{R}.$$

Exercise 3.

1. Let $n \geq 2$ be an integer. Prove that the polynomial $X^{2n} - nX^{n+1} + nX^{n-1} - 1$ has 1 as a triple root in $\mathbb{C}[X]$.
2. Let $P(X) \in \mathbb{C}[X]$. Find the multiplicity of the root α of the polynomial

$$\frac{X - \alpha}{2} (P'(X) + P'(\alpha)) - P(X) + P(\alpha).$$

3. Find the condition under which the polynomial $X^5 + aX^3 + b$ has a non-zero real double root.
4. Let n and m be two positive integers such that $n \geq m$ and let a, b be two real numbers. Prove that the polynomial $X^n + aX^{n-m} + b$ cannot have non-zero roots above multiplicity two.

Exercise 4. ★

Let $\mathbb{K}$ be a field and let $P_1(X), P_2(X), P(X), Q(X) \in \mathbb{K}[X]$.

1. Prove that $P_1(X) - P_2(X)$ divides $Q(P_1(X)) - Q(P_2(X))$.
2. Deduce that $P(X) - X$ divides $P(P(X)) - X$.
3. Deduce that, for all $k \in \mathbb{N}^*$: $P(X) - X$ divides $P^{[k]}(X) - X$, where $P^{[k]} = \underbrace{P \circ P \circ \dots \circ P}_{k \text{ times}}$.

Exercise 5. ★

1. For what values of the positive integer n does $X^2 + X + 1$ divide $(X + 1)^n + X + 1$ in $\mathbb{R}[X]$?
2. For what values of the positive integer n does $X^2 + X + 1$ divide $(X + 1)^n - X^n - 1$ in $\mathbb{R}[X]$?
3. Find a polynomial $P(X) \in \mathbb{R}[X]$ of degree 5 such that $P(X) - 1$ is divisible by $(X - 1)^3$ and $P(X)$ itself is divisible by X^3 .
4. Show that if $a, b, c, d \in \mathbb{N}$, then $X^3 + X^2 + X + 1$ divides $X^{4a+3} + X^{4b+2} + X^{4c+1} + X^{4d}$ in $\mathbb{R}[X]$.
5. Let $P(X), Q(X), R(X) \in \mathbb{C}[X]$ such that

$$P(X^3) + XQ(X^3) = (1 + X + X^2)R(X).$$

Show that $P(X)$, $Q(X)$, and $R(X)$ are all divisible by $X - 1$.

Exercise 6. ★

1. Using the Euclidean algorithm, find in $\mathbb{R}[X]$ the GCD of the following pairs of polynomials.
 - (a) $(P(X), Q(X)) = (X^5 - X^4 + 2X^3 + 1, X^5 + X^4 + 2X^2 - 1)$.
 - (b) $(P(X), Q(X)) = (nX^{n+1} - (n+1)X^n + 1, X^n - nX + n - 1)$, where $n \in \mathbb{N}^*$.
 - (c) $(P(X), Q(X)) = (X^n - 1, X^m - 1)$, where $(n, m) \in \mathbb{N}^* \times \mathbb{N}^*$.
2. In each of the following cases, show that $P(X)$ and $Q(X)$ are coprime in $\mathbb{R}[X]$. Then determine $S(X)$ and $T(X)$ such that $S(X)P(X) + T(X)Q(X) = 1$.
 - (a) $P(X) = X^3 - 2$ and $Q(X) = X^2 - 1$.
 - (b) $P(X) = X^3 + 1$ and $Q(X) = X^2 + X + 1$.
 - (c) $P(X) = X^4 - 1$ and $Q(X) = X^2 - 4$.

Exercise 7. ★

Let $n \in \mathbb{N}^*$. In each of the following cases, find the remainder of the Euclidean division of $P(X)$ by $Q(X)$ in $\mathbb{R}[X]$.

1. $P(X) = (X-1)^{n+2} + (X+2)^n - 2$ and $Q(X) = (X-1)^n$.
2. $P(X) = X^{3n} + 3X + 1$ and $Q(X) = (X-1)^2$.
3. $P(X) = (\cos a + (\sin a)X)^n$ and $Q(X) = X^2 + 1$, where $a \in \mathbb{R}$ and $n \geq 2$.
4. $P(X) = (X-3)^{2n} + (X-2)^n - 2$ and $Q(X) = (X-3)(X-2)$.
5. $P(X) = (X-3)^{2n} + (X-2)^n - 2$ and $Q(X) = (X-2)^2$.
6. $P(X) = X^{n+1} - X^n + 1$ and $Q(X) = X^2 - 1$.
7. $P(X) = X^n + X + 2$ and $Q(X) = X^3 + X^2 + X + 1$, where $n \geq 3$.
8. $P(X) = X^{5n}$ and $Q(X) = X^5 - 1$.

Exercise 8. ★(Factorization in $\mathbb{R}[X]$ and $\mathbb{C}[X]$ with applications)

1. Let n be a positive integer. Factor the following polynomials into irreducible real factors.

- | | |
|--------------------------------------|---|
| (a) $X^4 + 4$. | (g) $(X^2 - 2X + 1)^2 + 1$. |
| (b) $X^4 - 2X^3 + 9X^2 - 10X + 20$. | (h) $X^{2n} - 2X^n + 2$. |
| (c) $X^6 + 27$. | (i) $X^n - 1$. |
| (d) $X^8 - 1$. | (j) $X^n + 1$. |
| (e) $X^8 + 2X^6 + 3X^4 + 2X^2 + 1$. | (k) $1 - X + X^2 + \cdots + (-1)^n X^n$. |
| (f) $X^8 + X^4 + 1$. | |

2. (a) Factorize the polynomial $X^5 - 1$ in $\mathbb{R}[X]$ and in $\mathbb{C}[X]$.
- (b) Deduce $\cos \frac{2\pi}{5}$ and $\cos \frac{\pi}{5}$.

Exercise 9. ★

1. Write all polynomials of degree less or equal to 2 in $\mathbb{Z}/2\mathbb{Z}[X]$ and $\mathbb{Z}/3\mathbb{Z}[X]$. Denote these sets by A and B respectively.
2. Among these $P(X)$ polynomials determine the irreducible ones.
3. For any $P(X) \in A$, calculate $P(\bar{0})$ and $P(\bar{1})$.
4. Let $P \in \mathbb{Z}/2\mathbb{Z}[X]$, show that the polynomial map $P(X)$ is zero if and only if $P(X) \equiv 0 \pmod{X^2 + X}$ and is the identity if and only if $P(X) \equiv X \pmod{X^2 + X}$.
5. Let $P(X) \in \mathbb{Z}/2\mathbb{Z}[X]$ be irreducible. Show that the polynomial map $P(X)$ is the constant map equals to $\bar{1}$.

Exercise 10. ★(Applications of elementary symmetric functions)

1. Solve in $\mathbb{C}$ the following systems :

$$(a) \begin{cases} x + y + z = 0, \\ xy + yz + zx = -13, \\ xyz = -12. \end{cases}$$

$$(c) \begin{cases} x + y + z = -4, \\ \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = -\frac{1}{10}, \\ x^2 + y^2 + z^2 = 18. \end{cases}$$

$$(b) \begin{cases} x + y + z + t = -3, \\ xy + xz + yz + xt + yt + zt = 3, \\ xyz + xyt + xzt + yzt = -3, \\ xyzt = 2. \end{cases}$$

$$(d) \begin{cases} x + y + z = -1, \\ (x-y)^2 + (y-z)^2 + (z-x)^2 = \frac{7}{2}, \\ (x+2yz)^2 + (y+2xz)^2 + (z+2xy)^2 = \frac{27}{4}. \end{cases}$$

2. Find a polynomial of degree three whose zeros are the squares of the zeros of the polynomial $X^3 + 2X^2 - 3X + 5$.
3. Let $a, b, c, d \in \mathbb{C}$ such that $a \neq 0$. Find a necessary and sufficient condition for which the polynomial $X^4 + aX^3 + bx^2 + cx + d$ has two zeros of opposite signs.

Exercise 11. ★(Rational fractions with applications)

1. Let $n, m \in \mathbb{N}^*$. Split the following fractions into sums of partial fractions in $\mathbb{C}(X)$ and in $\mathbb{R}(X)$.

$$(a) \frac{1}{X^3 - X}.$$

$$(e) \frac{3}{(X^3 - 1)^2}.$$

$$(j) \frac{1}{X^2(X-1)^n}.$$

$$(b) \frac{X^2 + 2X + 5}{X^2 - 3X + 2}.$$

$$(f) \frac{X^2 + 1}{(X-1)^2(X^3 + 8)}.$$

$$(k) \frac{X^{2n}}{X^{2n+1} - 1}.$$

$$(c) \frac{X^3 + 1}{(X-1)^2}.$$

$$(g) \frac{1}{X^n - 1}.$$

$$(l) \frac{n!}{X(X-1)\cdots(X-n)}.$$

$$(d) \frac{X^5}{(X-2)^2(X-3)^2}.$$

$$(h) \frac{X^{n-1}}{X^n - 1}.$$

$$(i) \frac{X^5}{(X^2 - X + 4)^n}.$$

$$(m) \frac{1}{X^n(X-1)^m}.$$

2. In each case of the following, use a partial fractions representation of u_k to determine $\sum_{k=2}^n u_k$:

$$(a) u_k = \frac{1}{k(k-1)}.$$

$$(b) u_k = \frac{1}{k^3 - k}.$$

3. Using the division according to the increasing power order, split in $\mathbb{R}(X)$:

$$(a) \frac{4X^4 - 10X^3 + 8X^2 - 4X + 1}{X^3(X-1)^2}.$$

$$(b) \frac{X^4 + 2X^2 + 1}{X^5 - X^3}.$$

Exercise 12. (Polynomial with integral roots)

Let a be a positive integer and let $P_a(X) = X^3 - (a^2 + 2a)X + 2$. The goal of this exercise is to find all the values of a for which $P_a(X)$ has three integral roots.

1. Suppose that $P_a(X)$ has three roots $\alpha_1, \alpha_2, \alpha_3$ in $\mathbb{Z}$, where $\alpha_1 \leq \alpha_2 \leq \alpha_3$.

(a) What is the value of $\alpha_1 + \alpha_2 + \alpha_3$ and $\alpha_1\alpha_2\alpha_3$?

(b) Calculate $P_a(0)$, then deduce that $\alpha_1 < 0$.

(c) Deduce that $\alpha_1 \leq \alpha_2 \leq \alpha_3 \leq -\alpha_1$, then find the values of $\alpha_1, \alpha_2, \alpha_3$.

(d) Show that $P'_a(\alpha_2) = 0$. Deduce the value of a .

2. Conversely, show that the found value of a gives us a polynomial with integral roots.

Exercise 13.

Let $n \in \mathbb{N}^*$.

1. Show that there exists a unique $(P(X), Q(X)) \in (\mathbb{K}_{n-1}[X])^2$ such that $(1-X)^n P(X) + X^n Q(X) = 1$.
2. Show that $Q(X) = P(1-X)$ and $P(X) = Q(1-X)$.
3. Show that there exists $\lambda \in \mathbb{K}$ such that $(1-X)P'(X) - nP(X) = \lambda X^{n-1}$.

4. We write $P(X) = \sum_{k=0}^{n-1} a_k(X-1)^k$.

(a) Verify that $\lambda = -na_0$.

(b) Show that for all $k \in \{0, \dots, n-1\}$, $a_k = \frac{na_0}{n+k} \binom{n-1}{k}$.

(c) Show that $\sum_{k=0}^{n-1} (-1)^k a_k = 1$.

(d) Deduce $P(X)$.

Exercise 14. (Chebyshev polynomial)

We define a sequence of polynomials $(T_n(X))_{n \in \mathbb{N}}$ by setting : $T_0(X) = 1, T_1(X) = X$ and $T_{n+2}(X) = 2XT_{n+1}(X) - T_n(X)$ for all $n \in \mathbb{N}$. The polynomial $T_n(X)$ is called the n -th Chebyshev polynomial.

1. (a) Determine the degree of $T_n(X)$ and calculate its leading coefficient.

(b) Calculate the constant coefficient of $T_{2n}(X)$.

2. (a) Show that for all $\theta \in \mathbb{R}$:

$$T_n(\cos \theta) = \cos(n\theta). \quad (1)$$

(b) Show that $T_n(X)$ is the only polynomial of $\mathbb{R}[X]$ for which the relation (1) is true.

(c) By differentiating the relation (1) twice, show that $(X^2 - 1)T_n''(X) + XT_n'(X) - n^2T_n(X) = 0$.

3. Using the de Moivre's formula, determine an explicit expression of $T_n(X)$ for all $n \in \mathbb{N}$.

4. Let $n \geq 1$.

(a) Find all the roots of $T_n(X)$ that belong to $[-1, 1]$.

(b) Deduce that $T_n(X)$ is the product of linear factors over $\mathbb{R}$.

Quote taken from the book "**Geometry**" by Michèle Audin : "One must do exercises. One must seek out exercises. An exercise is not something of which one must know <<the>> solution to recite it to a jury. No concept can be understood or assimilated without a minimum of practice, research, and failures. **An exercise on which one has not <<struggled>> is a useless exercise.**"